The bad news is that this consistent picture of the Standard Model also tells us that our Higgs vacuum—the perfectly balanced set of laws that govern the physical world—is not stable.

Our whole beautiful cosmos appears to be living on borrowed time.

A SLIPPERY-SLOPE COSMOS

The idea that our vacuum might not be stable isn't a new one. Even in the 1960s and 1970s, physicists were gleefully writing papers imagining the possibilities for a universe that might undergo a catastrophic decay process, destroying all life as we know it and even all possibility of organized matter. Of course, at the time, vacuum decay was just a fun idea to play around with in the equations, with no experimental data to back it up.

Unlike now.

To understand vacuum decay, we have to understand the concept of a *potential*, a mathematical construct that represents how the value of a field can change, and where it "prefers" to be. You can think of the Higgs field as a pebble rolling down a slope into a valley, with the potential represented by the shape of that slope. Just as a pebble will settle in the bottom of a valley, the Higgs field will seek the lowest-energy state, where the potential is at its lowest value, and settle there, assuming nothing stops it. A sketch of the potential might look like a U shape, with the bottom of the U being the bottom of that valley. When electroweak symmetry breaking happened, it created the potential that the Higgs field is governed by, and as we generally imagine it, the Higgs is now safely settled at the bottom.

The problem is, that might not really be the bottom. There might be another vacuum state, at some even lower part of the potential. Imagine a kind of tilted rounded W shape, with one of the valleys, the one representing where our Higgs field doesn't live, a bit lower than the other. If the Higgs potential has that second, lower valley, it suddenly goes from being a nice mathematical construct to an existential threat to the cosmos.

Wherever the Higgs field is now in its potential, it's given us a perfectly livable, comfortable universe. We have constants of nature that are nicely compatible with bound particles and solid, life-compatible structures. If another state is possible, lower down the potential, all that is at risk.

In such a situation, the Higgs vacuum is only *meta*stable. Stable … ish … for now. The field is stuck in a part of the potential that looks like the